

DedeePYa Yarra

Machine Learning | MLOps Engineer

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Summary

- Results-driven Machine Learning Engineer with 4+ years of experience in MLOps, adept at designing and deploying machine learning models to solve complex business problems.
- Seeking an opportunity to leverage my MLOps, machine learning, and software engineering expertise to contribute to cutting-edge projects in a dynamic organization.
- Adept at leading technical teams, implementing real-time machine learning solutions, and leveraging Python programming skills to drive innovative projects.
- Skilled in Statistical Modeling, and Data Mining, with a focus on developing solutions for diverse business problems and generating data visualizations.
- Well-versed in developing different Statistical Machine Learning and Data Mining solutions to various business problems and generating data visualizations using R, Python, and MATLAB.
- Knowledge of building various machine learning models using algorithms such as Linear Regression, Logistic Regression, Naive Bayes, Support Vector Machines (SVM), Decision trees, and KNN Clustering.
- Extensive experience in designing and implementing supervised, unsupervised, and reinforcement learning models, as well as deep learning techniques, to solve complex business challenges and enhance decision-making through advanced predictive analytics.
- Good knowledge of Python and R packages and libraries like ggplot2, Pandas, NumPy, SciPy, Matplotlib, Seaborn, TensorFlow, and Scikit-learn.
- Strong understanding of AWS services (S3, EC2, Redshift) and a passion for AI advancements, particularly in Generative AI and Large Language Models, seeking a challenging role in AI research and development.
- Experienced in data preprocessing, feature selection, and data visualization techniques to create high-performing ML models.

Skills

Methodology: SDLC, Agile, Waterfall.

Programming Language: Python, R, SQL, MATLAB, Spark, Java, Jira, PowerShell, C#.

Machine Learning: Linear Regression, Logistic Regression, Random Forest, Association Rules, Support Vector Machine, Naive Bayes, k -Means, Clustering, Classification, Gradient Boosting Trees, PCA, LDA, K- Nearest Neighbors(KNN), Natural Language Processing, A/B Testing, Statistics, Information retrieval, Ads, Search engine, Recommender System, CTR/CVR model estimation accuracy, data analysis, modeling, feature engineering.

MLOps Tools: Apache Airflow, Kubernetes, Docker, Jenkins, Model deployment, Databricks, Datadog, CI/CD Pipelines.

Deep Learning: Artificial Neural Networks, Convolutional Neural Networks, RNN, Deep Learning on AWS, Keras API, Transforms, images, GANS, Generative AI, Large Language Models (LLMs), Diffusion Models, ChatGPT, BERT, Fine Tuning, Prompt Design, Prompt Engineer, Vertex AI, Conversational Engineering, Conversational AI Development, Natural Language Processing, Computer Vision, Computer Graphics, Simulation, NLP, Ranking, query classification, seq2seq, NER (Named Entity Recognition), knowledge graph, bid word optimization,

Data Visualization: Tableau, Tableau server, Tableau Reader, PowerBI

Packages: ggplot2, Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Keras, Spark, Matplotlib, SciPy.

Big Data Technologies: Hadoop, Spark, Kafka, MapReduce, Hive QL, HDFS, Sqoop, Pig.

ETL Tools: Informatica Power Centre SSIS, Apache Airflow, Jupyter, Rapid Miner

Database: MySQL, SQL Server, MongoDB, HBase, Impala Amazon Redshift, Teradata.

Operating Systems: Windows, Linux

Other Tools and technologies: Git, MS Office, AWS Sagemaker , Azure ML studio, GCP, NLTK, Gensim, Google Analytics, AWS-(EC2/S3/Redshift/EMR/Lambda/Snowflake), Azure (VMs/Blob Storage/Data Factory/Synapse/Functions/Databricks).

Experience

Ebay, TX | March 2023 - Current | Machine Learning / MLOps Engineer

- Implementing Agile development methodology within the cross-functional team and acting as a liaison between the business user group and the technical team.
- Developed, implemented, and managed scalable operations research and machine learning pipelines for data preprocessing, model training, validation, and deployment on Databricks and Gurobi Cloud using Terraform.
- Collaborated closely with data scientists and software engineers to deploy machine learning models into production, optimizing their performance for real-world scenarios.
- Built a robust monitoring and alerting system for deployed machine learning models, utilizing tools such as Datadog and custom monitoring scripts to ensure real-time tracking of model performance and health, which resulted in a 30% reduction in downtime and improved model accuracy through faster issue resolution.
- Applied NLP techniques to analyze customer feedback, leveraging AWS services for large-scale processing and driving the development of new features that improved user retention.

- Designed and deployed machine learning models for various applications like Natural Language Processing, and recommendation systems using Python and SQL to enhance product recommendations, resulting in a 15% increase in customer engagement and conversion rates.
- Developed and maintained large-scale data pipelines on AWS, preprocessing millions of e-commerce transactions for demand forecasting and inventory management, ensuring high data accuracy and consistency.
- Implemented automated model deployment and monitoring using Docker, Kubernetes, and MLOps best practices, reducing model deployment time by 40% and ensuring continuous monitoring to address data drift.
- Optimized fraud detection models by collaborating with data scientists and utilizing statistical techniques, reducing false positives by 20% and enhancing transaction security.
- Conducted advanced data analytics using Python and SQL to extract actionable insights from e-commerce data, supporting decision-making processes across multiple teams.
- Integrated real-time machine learning models into eBay's search algorithm, improving search relevance and user satisfaction by 10%, while scaling solutions to handle high transaction volumes with minimal latency.
- Implemented A/B testing, statistical validation and customer segmentation techniques, resulting in data-driven decisions and a 15% improvement in the retention rate of high-value customers.
- Worked with cloud-native technologies like Azure functions, Synapse Analytics, AWS S3, RDS, and Lambda to build scalable machine learning infrastructure, reducing operational costs and improving model performance.
- Automated model monitoring and alerting systems to track data drift and model performance, ensuring timely interventions and maintaining model accuracy.
- Optimized large-scale data processing and model training using Databricks and Snowflake, improving data pipeline efficiency by 30% and enhancing model performance.
- Conducted data preprocessing, feature engineering, and model evaluation to optimize model performance.
- Led efforts to improve model interpretability and explainability, enabling better decision-making and compliance with regulatory requirements.
- Collaborated with data engineers to create data pipelines, ensuring data quality and reliability for machine learning tasks.
- Conducted regular knowledge-sharing sessions to educate team members on MLOps best practices, fostering a culture of continuous learning and innovation.

Byte Work Solutions, India | Jan 2020 - July 2022 | Machine Learning / MLOps Engineer

- Worked in Waterfall Methodology and was involved in the project discussions.
- Led efforts to develop and maintain MLOps pipelines, enabling seamless integration of machine learning models into the production environment.
- Deployed and managed containerized machine learning models using Kubernetes and Docker, ensuring scalable and reliable operations in production environments.
- Collaborated with cross-functional teams to ensure successful deployment and optimization of machine learning models for real-world applications.
- Developed custom Python scripts and utilized SQL for data extraction, transformation, and validation, ensuring clean, accurate data for high-performing models in production.
- Conducted regular performance evaluations, identified areas for improvement, and recommended enhancements to streamline MLOps workflows.
- Developed machine learning algorithms to analyze large datasets and extract actionable insights for clients in various industries.
- Built and maintained data pipelines for data ingestion, transformation, and storage, enhancing data accessibility and usability.
- Worked on text classification, sentiment analysis, and anomaly detection projects, that engaged over 5,000 customer interactions, providing in-depth reports that informed product development, resulting in three new feature launches directly addressing user feedback.
- Contributed to research and development efforts, staying up to date with the latest advancements in machine learning and MLOps.

Certifications

- **Microsoft Certified DevOps Engineer Expert.**
- **Microsoft Azure Administrator Associate.**
- **Source control Git and GitHub.**
- **Python Essentials.**
- **Power BI Essentials.**

Education Details

Master of Science in Data Science
University of North Texas, Denton, TX

Aug 2022 - Dec 2023

Bachelor of Technology in Computer Science and Engineering
Jawaharlal Nehru Technological University, Kakinada, Andhra Pradesh

June 2016 - Sep 2020